05.06.2024

**EE2004 MICROPROCESSORS FINAL EXAM**

**Name:**

**CLOSED BOOK, CLOSED NOTES EXAM. NO CALCULATORS OR CHEAT SHEETS ALLOWED.**
**PLEASE PROVIDE YOUR ANSWERS IN DESIGNATED PLACES.**
**DURATION: 60 min.**

**Q1) (20 points)** A DAC is used to output a sine wave using SysTick Interrupts and a sine-wave table. Assume the DAC has 8 bits, the DAC output is connected to a speaker, the SysTick ISR executes at 100kHz. How many elements should the sine table include in order to produce a 400 Hz sound?

**Q2) (20 points)** Three microcontrollers (A, B & C) have maximum clock speeds 10MHz, 20MHz, 24MHz respectively. Microcontroller A divides its clock by 4 to give one machine cycle, microcontroller B by 8, and microcontroller C by 12. Microcontrollers A & C take 2 machine cycles to perform an instruction, while microcontroller B takes three cycles. Place the microcontrollers in order of the speed in which they can perform that instruction?

**Q3) (10 points)** You have a 12-bit, 0 to 3V range ADC. If the ADC input is 0.75 V, what will be the digital value in hex returned by this ADC?

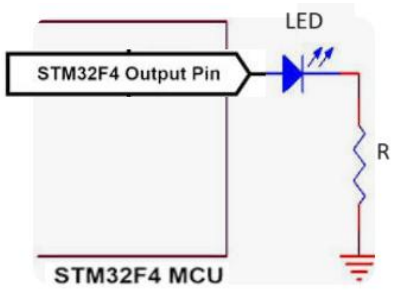
**Q4) (20 points)** When using the system timer, there is just one callback function named HAL_IncTick(void) for SysTick. This function is called every millisecond.
The aim of the code is blinking an LED with each 2 second intervals.

- Fill question marks (there are two) in the listing below.

```
/* USER CODE BEGIN PV */
uint16_t sysTickCnt = 0;
/* USER CODE END PV */

/* USER CODE BEGIN 0 */
void HAL_IncTick(void)
{
sysTickCnt++;
if (sysTickCnt >= ??  )
{
HAL_GPIO_TogglePin(GPIOG, GPIO_PIN_13);
sysTickCnt = ??;
}
}
/* USER CODE END 0 */
```

**Q5) (10 points)** Interface an LED to a selected output pin as given in the figure. The desired LED operating point is 2.4V, 1mA. The microcontroller output high voltage is 3.3V. Find required resistor value R.




**Q6) (20 points)** A DAC is used to output a sine wave using SysTick Interrupts and a sine-wave table. Assume the DAC has 7 bits, the DAC output is connected to a speaker, the SysTick ISR executes at 32kHz, the sine table has 256 elements, and one DAC output occurs each interrupt. The DAC output range is 0 to 3.3V. The bus clock is 80 MHz. The ADC maximum rate is 125 kHz. What frequency sound is produced, in Hz?